

CSE111: Computational Thinking, Systems & Career Readiness

100-Percentile Master Study Guide, Textbooks Deep-Dive & Curated Video Hub

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1. Standard Reference Textbooks & Deep-Dive Chapter Context

Unit	Subject	Standard Textbook	Core Chapter Concept & Exam Takeaways
Unit I	Operating Systems	Silberschatz, Galvin & Gagne (10th Ed)	Ch 1: Dual-Mode CPU execution (User Mode Ring 3 vs Kernel Mode Ring 0), Hardware Interrupts vs Traps. Ch 2: Batch, Multitasking (time quanta), Hard RTOS (deterministic deadlines) vs Soft RTOS. Ch 13: Inode hierarchy, octal permissions (chmod 4-2-1 math).
Unit I	Computer Networks	Kurose & Ross / Forouzan	Ch 1: Packet vs Circuit Switching, Mesh topology formula $N(N-1)/2$. Ch 5/6: Collision vs Broadcast domains, 48-bit MAC vs 32-bit IP, Layer 1 Hubs (dumb repeaters), Layer 2 Switches (MAC learning table), Layer 3 Routers (IP routing), Gateways.
Unit I	Cloud & Hypervisors	Thomas Erl (Cloud Computing)	Ch 4/5: SPI responsibility matrix (IaaS vs PaaS vs SaaS), Public/Private/Hybrid. Ch 11: Hypervisors: Type-1 Bare-Metal (ESXi/KVM on silicon, zero OS overhead) vs Type-2 Hosted (VirtualBox on host OS).
Unit II	Version Control	Scott Chacon (Pro Git, 2nd Ed)	Ch 1/2/3: Centralized (SVN) vs Distributed (Git DAG), 4-Stage State Machine (Working Dir → Staging → Local Repo → Remote GitHub), SHA-1 immutability, 3-Way Merge vs Fast-Forward.
Unit II	Cyber Security	William Stallings (Crypto & NetSec)	Ch 1: CIA Triad. Ch 22: Malware Taxonomy (Viruses require host, Worms self-replicate on networks, Trojans disguise backdoors, Ransomware locks keys, Rootkits hook Kernel Ring 0, Botnets DDoS). Ch 23: 3-Factor MFA, Least Privilege (PoLP).
Unit IV	AI & Machine Learning	Russell & Norvig (AIMA 4th Ed)	Ch 1/19/28: AI $\supset$ ML $\supset$ DL; Supervised (labeled X,Y) vs Unsupervised (clustering/PCA) vs Reinforcement Learning (Agent/State/Reward); Neural Networks (Weights, Biases, ReLU); GenAI vs Agentic AI (perception-tool-action loops); Prompt Engineering (CoT).
Unit V/VI	Career & Resume	Cracking the Coding Interview	Ch 1 & 2: Career Pathways (Product vs Service vs GATE vs Research); 5-Step IDP (Self-Assessment → Gap → SMART Goals → Action → Review); Single-page ATS Dream CV formatted via the STAR Method (Situation, Task, Action, Result) with bolded metrics.

2. Curated YouTube Video Playlists & Online Resource Links

Unit / Topic	Recommended Channel	Search Query / Video Reference	Core Takeaway & Concept
Unit I: Computational Thinking	Harvard CS50	CS50 2024 - Lecture 0 - Computational Thinking	4 Pillars: Decomposition, Pattern Recognition, Abstraction, Algorithms.
Unit I: OS Types & Architectures	Gate Smashers	Types of Operating System - Gate Smashers	Batch, Time-sharing, Distributed, and Hard/Soft RTOS.
Unit I: Linux CLI & Hierarchy	freeCodeCamp	Linux Command Line for Beginners Full Course	Root hierarchy, chmod 755 math, grep, mkdir -p.
Unit I: Network Devices	PowerCert Animated	Network Devices Explained - Hub, Switch, Router	Visual OSI Layer 1, Layer 2, and Layer 3 data routing.
Unit I: Hypervisors Type 1 & 2	NetworkChuck	NetworkChuck Hypervisor Type 1 and Type 2	Bare-Metal (ESXi) vs Hosted (VirtualBox) virtualization.
Unit II: Git & GitHub Masterclass	Kunal Kushwaha	Git and GitHub Tutorial for Beginners Kunal	Git 4-stage lifecycle, branching, pull requests.
Unit II: Malware Taxonomy	Simplilearn	Types of Malware Explained Simplilearn	Viruses, Worms, Trojans, Ransomware, Rootkits, Botnets.
Unit IV: Machine Learning	StatQuest	StatQuest Machine Learning Fundamentals	Supervised vs Unsupervised vs Reinforcement Learning.
Unit IV: Generative vs Agentic AI	Fireship / Andrew Ng	AI Agents in 100 Seconds Fireship	Perception-tool-action loops vs static GenAI.
Unit V/VI: ATS Resume Design	Kunal Kushwaha	How to write a software engineer resume	STAR formula bullet points with quantified metrics.

3. High-Yield Unit Summaries, Diagrams & Formulas

Unit I (Computing Systems): The 4 Pillars of Computational Thinking are *Decomposition, Pattern Recognition, Abstraction, and Algorithm Design*. Linux file permissions use octal math: Read (4), Write (2), Execute (1) — e.g. chmod 755 gives User (rwx), Group (r-x), Others (r-x). Network topologies include Star, Bus, Ring, and Mesh with $N(N-1)/2$ links. Hypervisors are Type-1 (Bare Metal: ESXi) and Type-2 (Hosted: VirtualBox). Cloud SPI tiers are IaaS, PaaS, and SaaS.

Unit II (Git & Security): Git 4 stages: Working Directory → git add → Staging Area → git commit → Local Repo → git push → Remote. Malware taxonomy: Virus (parasitic host required), Worm (autonomous network replication), Trojan (disguised backdoor), Ransomware (crypto extortion), Rootkit (Kernel Ring 0 compromise), Botnet (C2 zombies). 3-Factor MFA: Knowledge (password), Possession (OTP/token), Inherence (biometrics).

Unit III (EDU-Revolution): SCRGM (Student-Centric Revenue Generation Model) enables commercial freelancing/startups. RPL (Recognition of Prior Learning) converts hackathons (SIH) and certifications into Grade 'O' (10.0 CGPA) via UMS Standing Committee.

Unit IV (AI & Emerging Tech): AI contains ML which contains DL. 3 ML Paradigms: Supervised (Classification/Regression), Unsupervised (Clustering/PCA), Reinforcement Learning (Agent/State/Reward). GenAI is static request-response; Agentic AI runs autonomous perception-tool-action loops with native code execution. Emerging tech: Data Science 5Vs, IoT, Blockchain (Smart Contracts/PoS), Edge Computing, Quantum (Qubits/Superposition).

Unit V & VI (Career & Dream CV): Career Pathways (Product vs Service vs GATE/PSU vs Research vs Startups). 5-Step IDP (Self-Assessment → Skill Gap → SMART Goals → Action Plan → Review). ATS Dream CV uses the **STAR Method** (Situation, Task, Action, Result) with bolded metrics.

4. 100-Percentile Exam Presentation Blueprint

Tier 1: Formal Definition (1-2 lines with technical keywords bolded)
Tier 2: Labeled Architectural Block Diagram / Flowchart (Drawn neatly in a box)
Tier 3: Structured Comparative Table / Bullet Points (Highlighting contrast & parameters)
Tier 4: Real-World Example / Code Syntax (Demonstrating practical implementation)